

Project: Vehicle Performance Dashboard – Toyota Motors Europe

Overview:

At Toyota Motors Europe, I developed an interactive data dashboard to analyze and visualize key vehicle performance parameters. The tool was designed to support design and engineering teams with data-driven insights for better decision-making.

As this project was developed and deployed within Toyota's internal systems, I do not have a copy of the dashboard. However, I will walk you through the structure and core features as below:

Design decisions

- Focused on intuitive, interactive visualizations that can be used by both technical and non-technical stakeholders.
- Prioritized flexibility by allowing users to select vehicle parameters dynamically (ex. , curb weight, horsepower, torque, fuel efficiency, etc).
- Included scatterplots for relationship analysis (ex., acceleration vs horsepower, torque vs payload).
- Designed with filtering capabilities by vehicle type and resolution to enable detailed insights.
- Maintained and enhanced the dashboard based on changing team requirements.

Tools

- Streamlit is used for building the dashboard UI.
- Plotly was used to generate rich, interactive scatterplots and graphs.
- Included functionality to upload new data files, enabling data analysis without developer intervention.

Core features

- 1) Upload and process new datasets.
- 2) Interactive filtering by vehicle type and axis resolution.
- 3) Built a data parameter selection interface for customizing visuals.
- 4) Visualization of relationships of selected data parameters (scatterplots). Few examples are as below:
 - a) Curb Weight vs Fuel Efficiency
 - b) Acceleration vs Horsepower
 - c) Torque vs Payload
 - d) Wheelbase vs Turning Radius
 - e) Battery Capacity vs Range (EVs)

- f) Acceleration vs Safety Ratings
- 5) Identification of trends across manufacturing years and detection of clusters or outliers among vehicle models.

Impact:

The tool significantly improved the team's ability to:

- Explore and validate vehicle performance relationships interactively.
- Helped to gain important insights on parameter dependencies without requiring deep technical expertise.
- Improved the speed and depth of internal vehicle performance analysis.